First-Order Logic

Chapter 8

Examples

[04PBT 04PFT 04PLT 04PNT 04PRT]

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[dks0638]

First-order logic

- Whereas propositional logic assumes the world contains facts,
- first-order logic (like natural language) assumes the world contains
  - Objects: people, houses, numbers, colors, baseball games, wars, …
  - Relations: red, round, prime, brother of, bigger than, part of, comes between, …
  - Functions: father of, best friend, one more than, plus, …

Syntax of FOL

- Constants KingJohn, 2, UofA,...
- Predicates Brother, >,...
- Functions Sqrt, LeftLegOf,...
- Variables x, y, a, b,...
- Connectives $\neg$, $\Rightarrow$, $\wedge$, $\vee$, $\Leftrightarrow$
- Equality =
- Quantifiers $\forall$, $\exists$

Example Knowledge Base

- The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American.
- Prove that Col. West is a criminal!

Example Knowledge Base contd.

. . . it is a crime for an American to sell weapons to hostile nations:

$$American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z) \Rightarrow Criminal(x)$$

Nono . . . has some missiles, i.e., $\exists\, x\; Owns(Nono, x) \wedge Missile(x)$:

$\boxed{Owns(Nono, M_1)}$ and $\boxed{Missile(M_1)}$

. . . all of its missiles were sold to it by Colonel West

$$\forall\, x \;\; Missile(x) \wedge Owns(Nono, x) \;\Rightarrow\; Sells(West, x, Nono)$$

Missiles are weapons:

$$Missile(x) \Rightarrow Weapon(x)$$

An enemy of America counts as "hostile":

$$Enemy(x, America) \;\Rightarrow\; Hostile(x)$$

West, who is American . . .

$\boxed{American(West)}$

The country Nono, an enemy of America . . .

$\boxed{Enemy(Nono, America)}$

Forward chaining proof

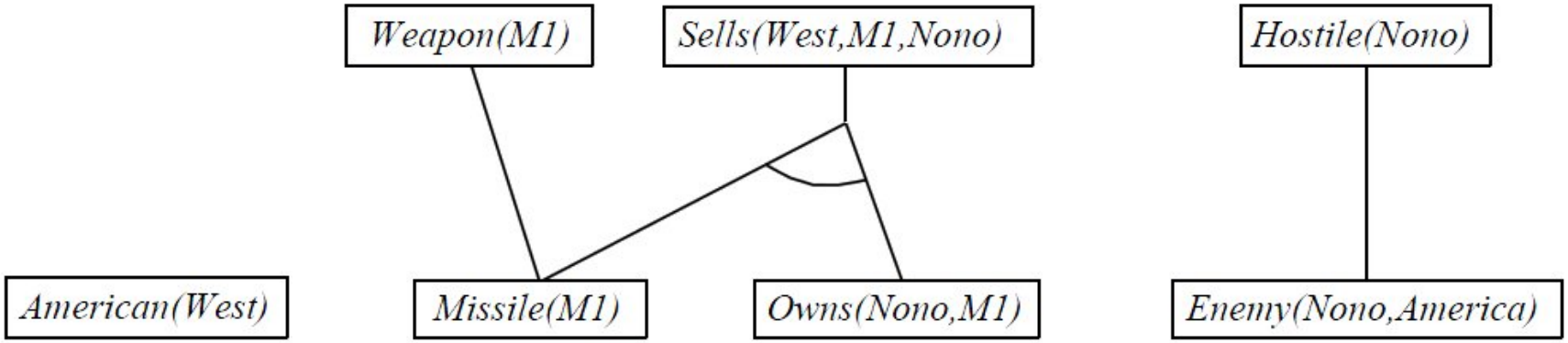
American(West)

Missile(M1)

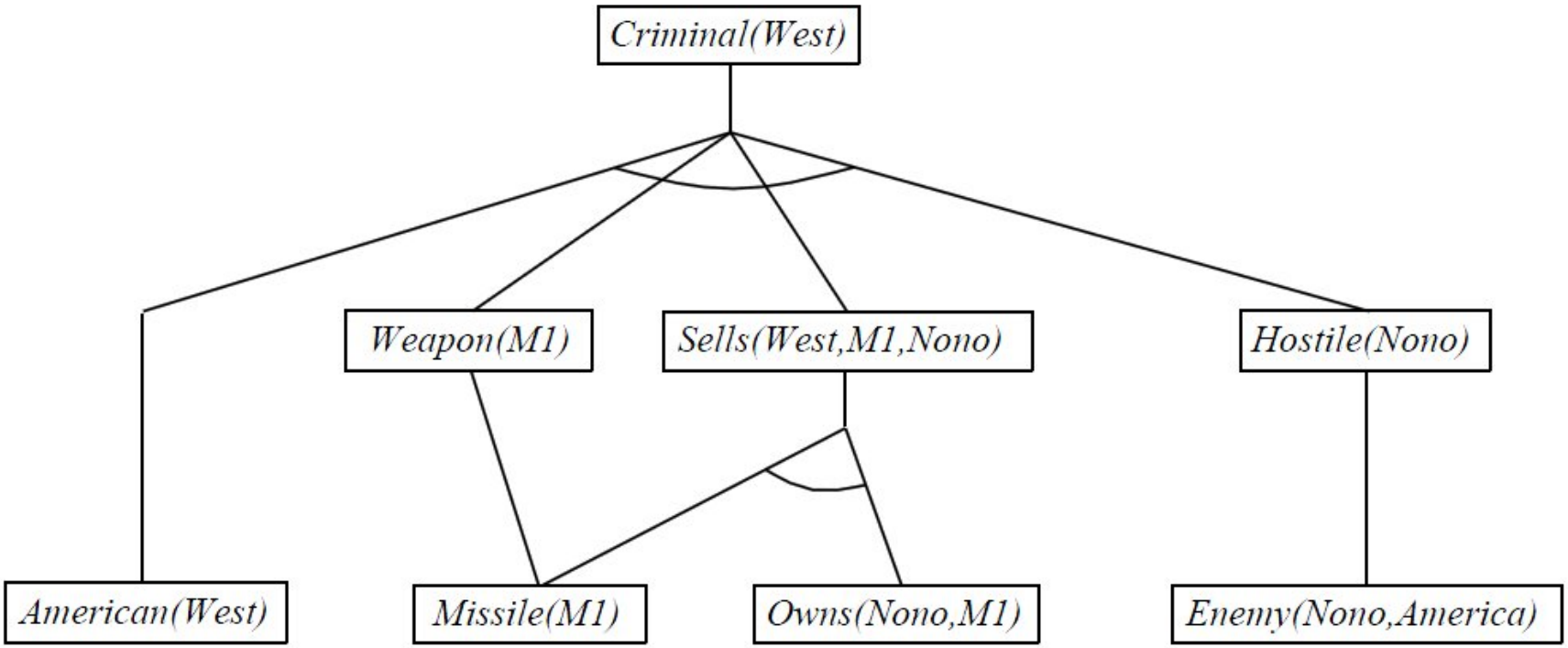
Owns(Nono,M1)

Enemy(Nono,America)

Forward chaining proof



Forward chaining proof



Properties of forward chaining

Sound and complete for first-order definite clauses (proof similar to propositional proof)

Datalog = first-order definite clauses + **no functions** (e.g., crime KB)

FC terminates for Datalog in poly iterations: at most $p \cdot n^k$ literals

May not terminate in general if α is not entailed

This is unavoidable: entailment with definite clauses is semidecidable

Efficiency of forward chaining

Simple observation: no need to match a rule on iteration *k* if a premise wasn't added on iteration *k - 1*

- match each rule whose premise contains a newly added literal

Matching itself can be expensive

Database indexing allows $O(1)$ retrieval of known facts
e.g., query *Missile(x)* retrieves $Missile(M_1)$

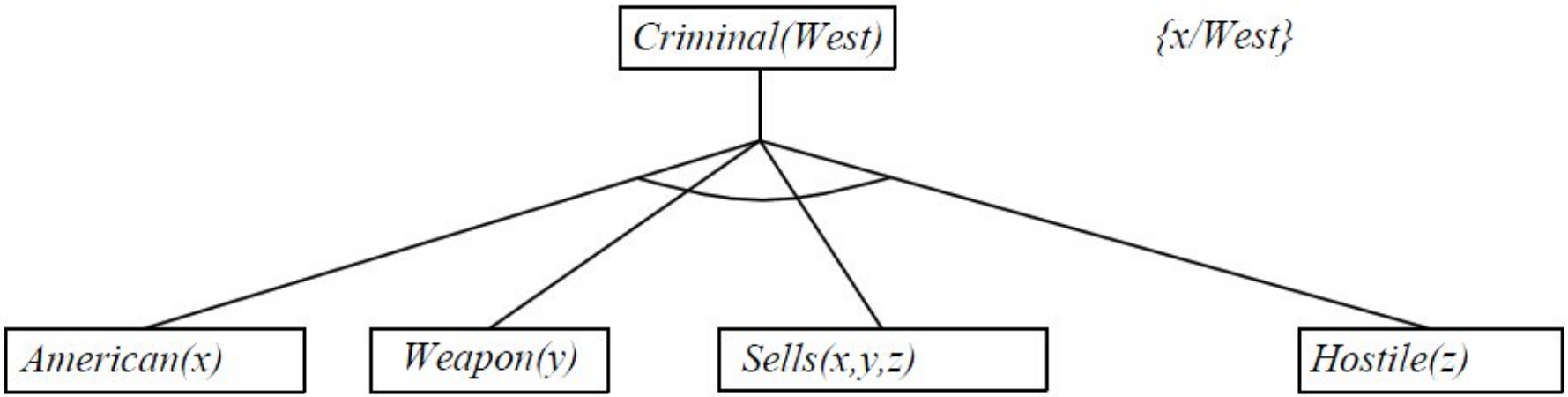
Matching conjunctive premises against known facts is NP-hard

Forward chaining is widely used in deductive databases

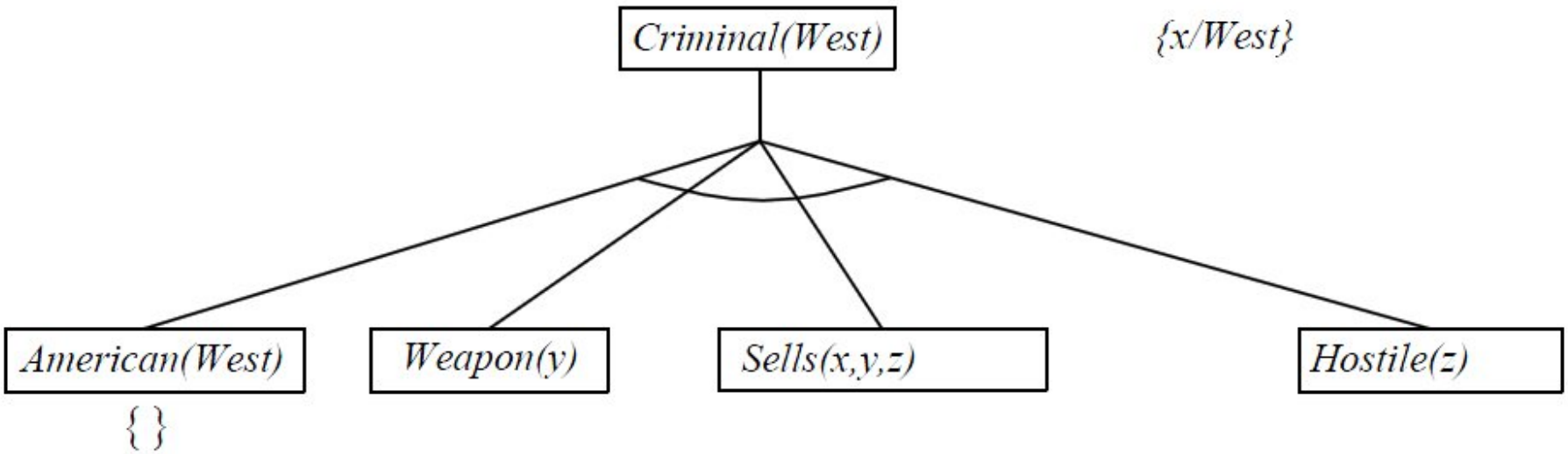
Backward chaining example

Criminal(West)

Backward chaining example



Backward chaining example



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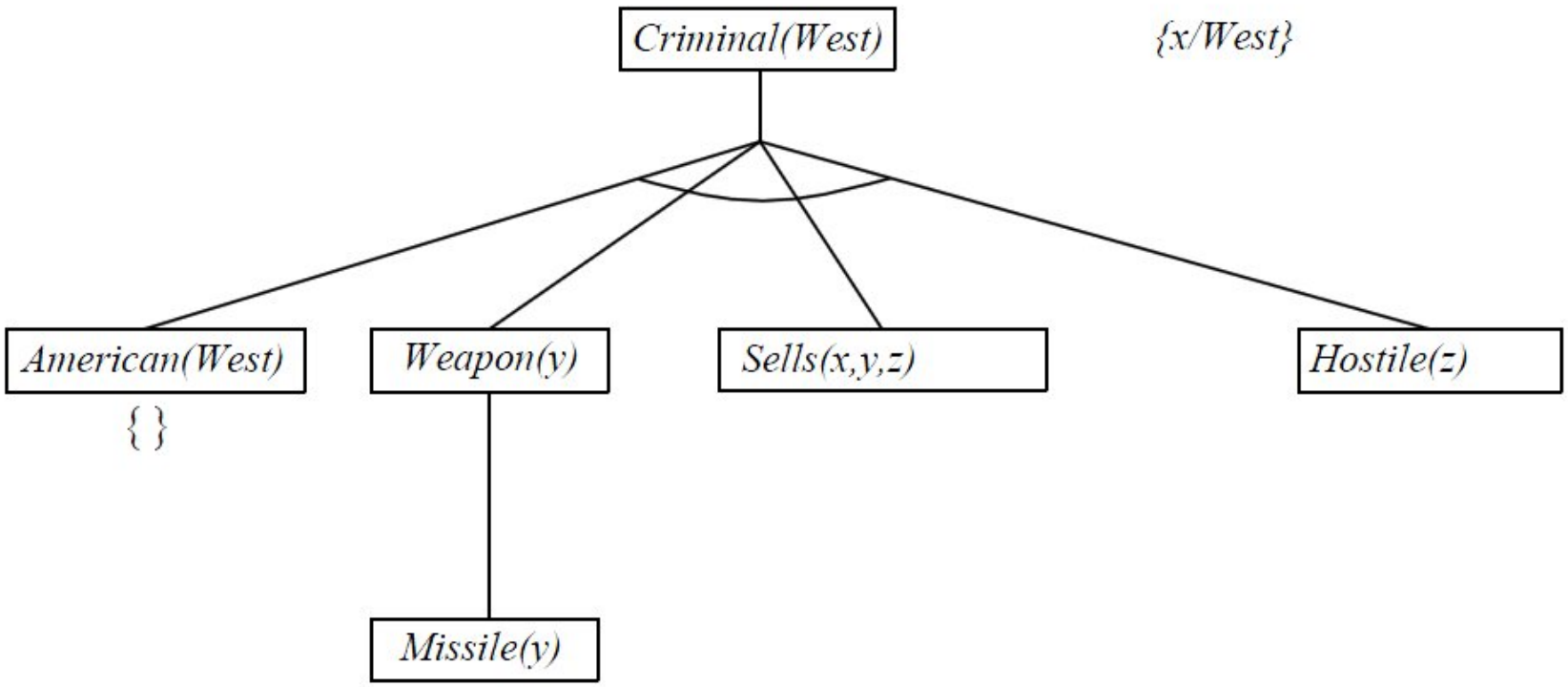

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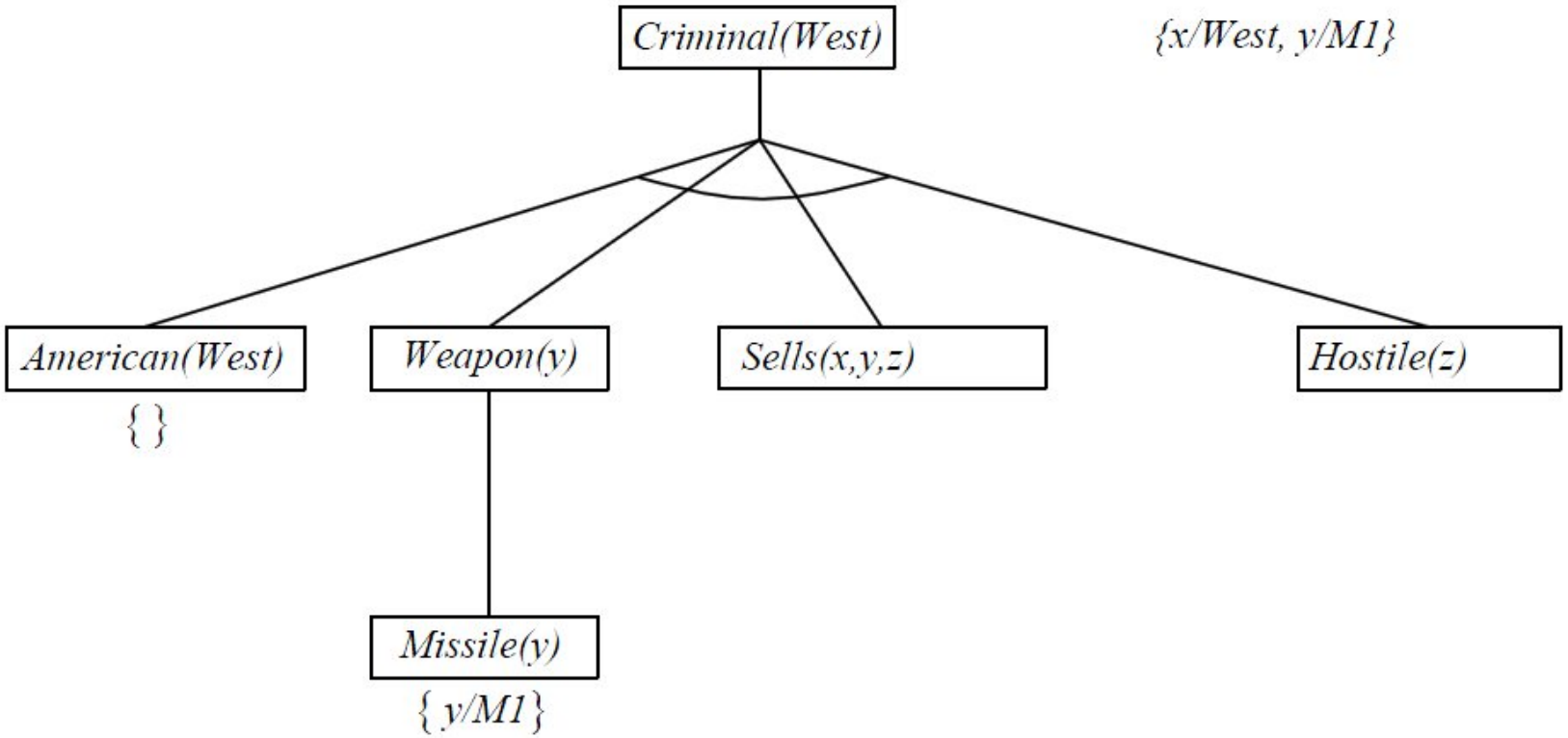

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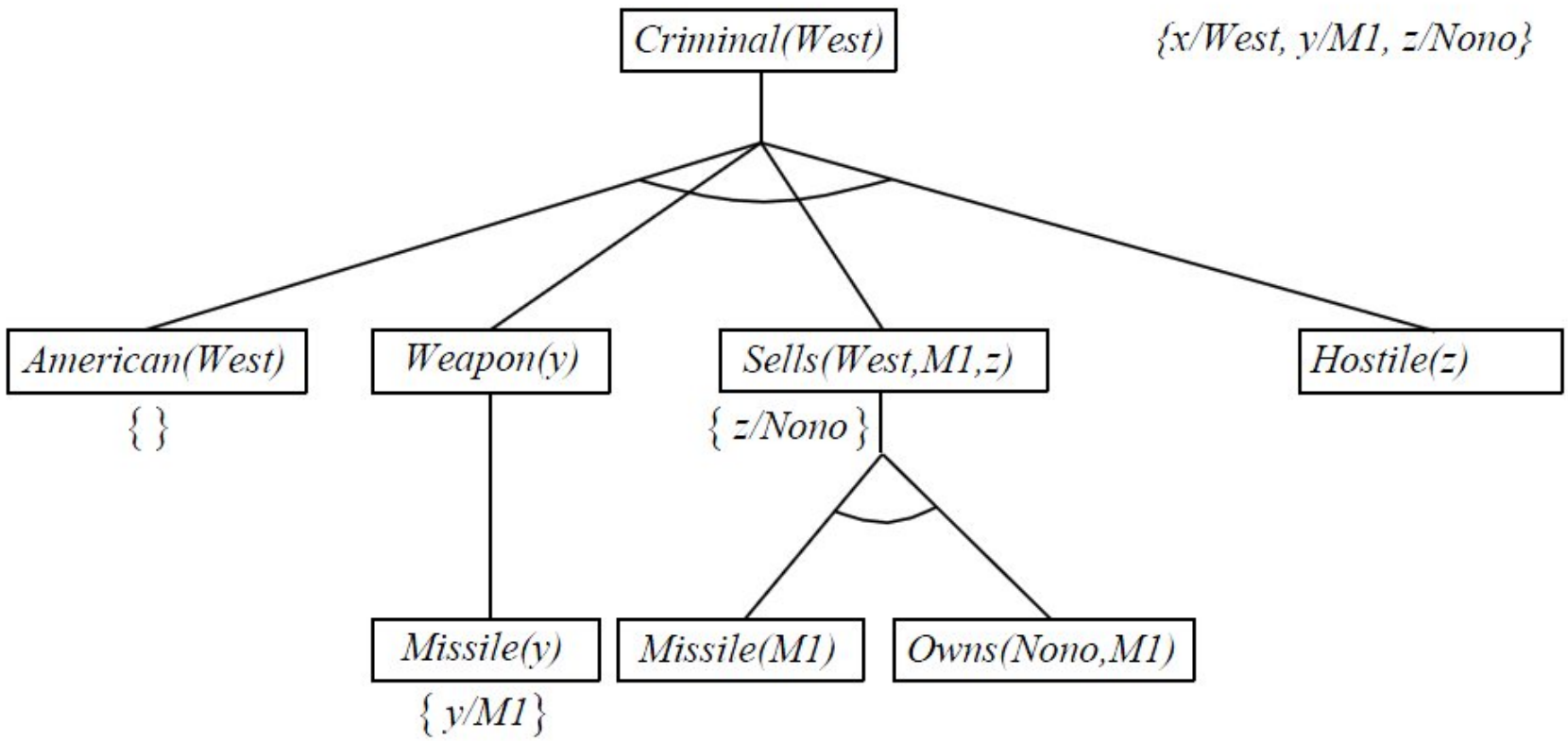

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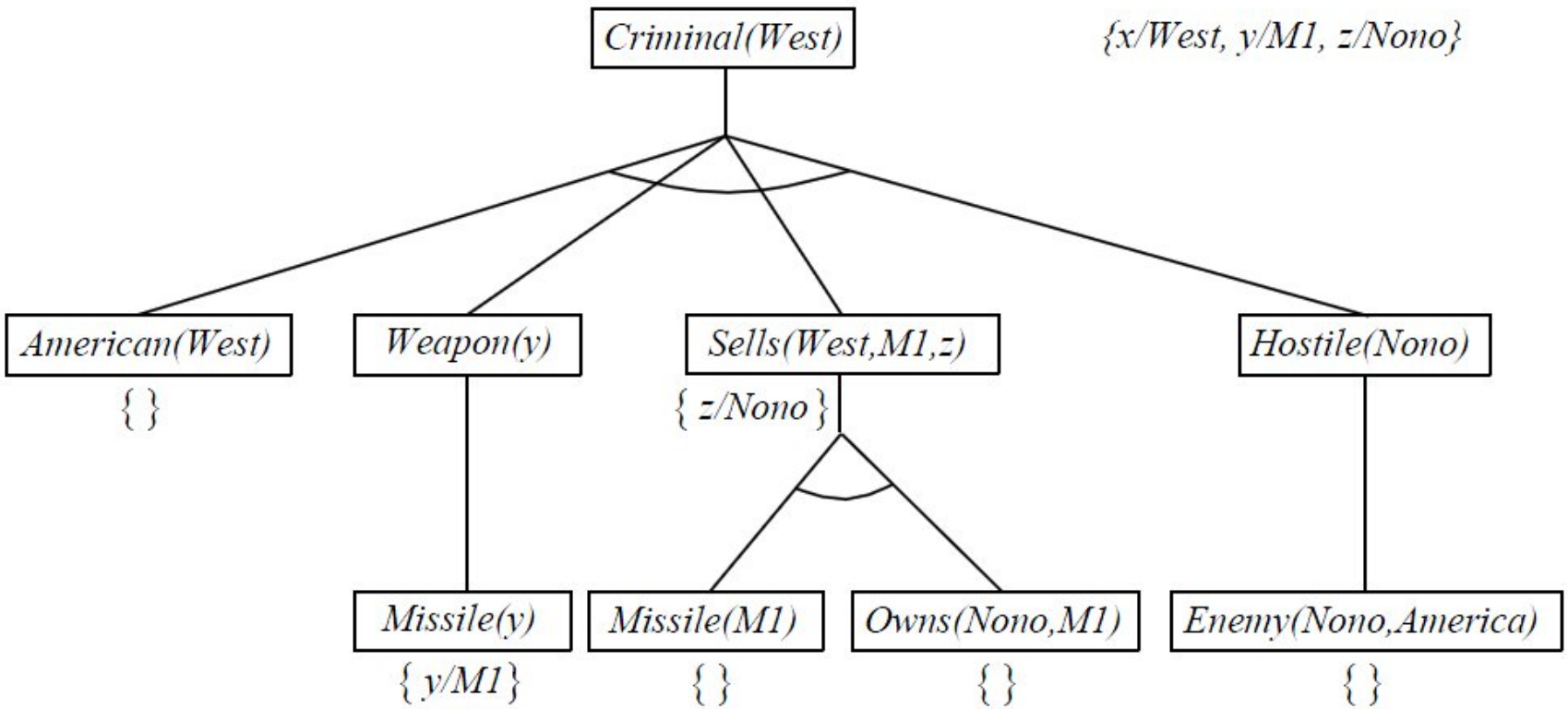


Properties of backward chaining

Depth-first recursive proof search: space is linear in size of proof

Incomplete due to infinite loops

$\Rightarrow$ fix by checking current goal against every goal on stack

Inefficient due to repeated subgoals (both success and failure)

$\Rightarrow$ fix using caching of previous results (extra space!)

Widely used (without improvements!) for logic programming

Conversion to CNF

Everyone who loves all animals is loved by someone:

$$\forall x \ [\forall y \ Animal(y) \Rightarrow Loves(x,y)] \Rightarrow [\exists y \ Loves(y,x)]$$

1. Eliminate biconditionals and implications

$$\forall x \ [\neg\forall y \ \neg Animal(y) \vee Loves(x,y)] \vee [\exists y \ Loves(y,x)]$$

2. Move $\neg$ inwards: $\neg\forall x, p \equiv \exists x \ \neg p$, $\neg\exists x, p \equiv \forall x \ \neg p$:

$$\forall x \ [\exists y \ \neg(\neg Animal(y) \vee Loves(x,y))] \vee [\exists y \ Loves(y,x)]$$
$$\forall x \ [\exists y \ \neg\neg Animal(y) \wedge \neg Loves(x,y)] \vee [\exists y \ Loves(y,x)]$$
$$\forall x \ [\exists y \ Animal(y) \wedge \neg Loves(x,y)] \vee [\exists y \ Loves(y,x)]$$

3. Standardize variables: each quantifier should use a different one

$$\forall x \;\; [\exists y \;\; Animal(y) \wedge \neg Loves(x, y)] \vee [\exists z \;\; Loves(z, x)]$$

4. Skolemize: a more general form of existential instantiation.
   Each existential variable is replaced by a Skolem function
   of the enclosing universally quantified variables:

$$\forall x \;\; [Animal(F(x)) \wedge \neg Loves(x, F(x))] \vee Loves(G(x), x)$$

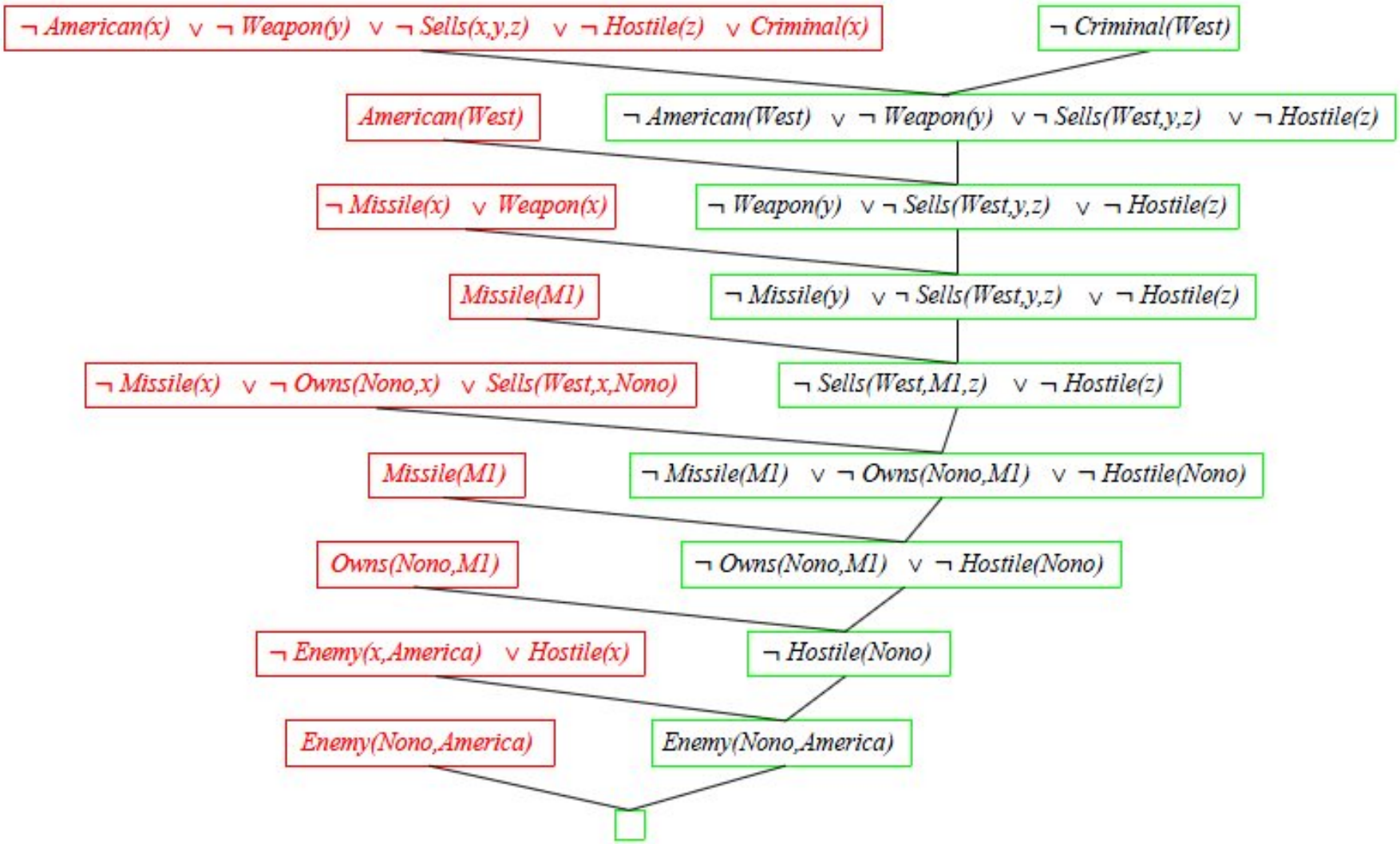
5. Drop universal quantifiers:

$$[Animal(F(x)) \wedge \neg Loves(x, F(x))] \vee Loves(G(x), x)$$

6. Distribute $\wedge$ over $\vee$:

$$[Animal(F(x)) \vee Loves(G(x), x)] \wedge [\neg Loves(x, F(x)) \vee Loves(G(x), x)]$$

Resolution proof: definite clauses



Resolution refutation proofs involves the following steps:

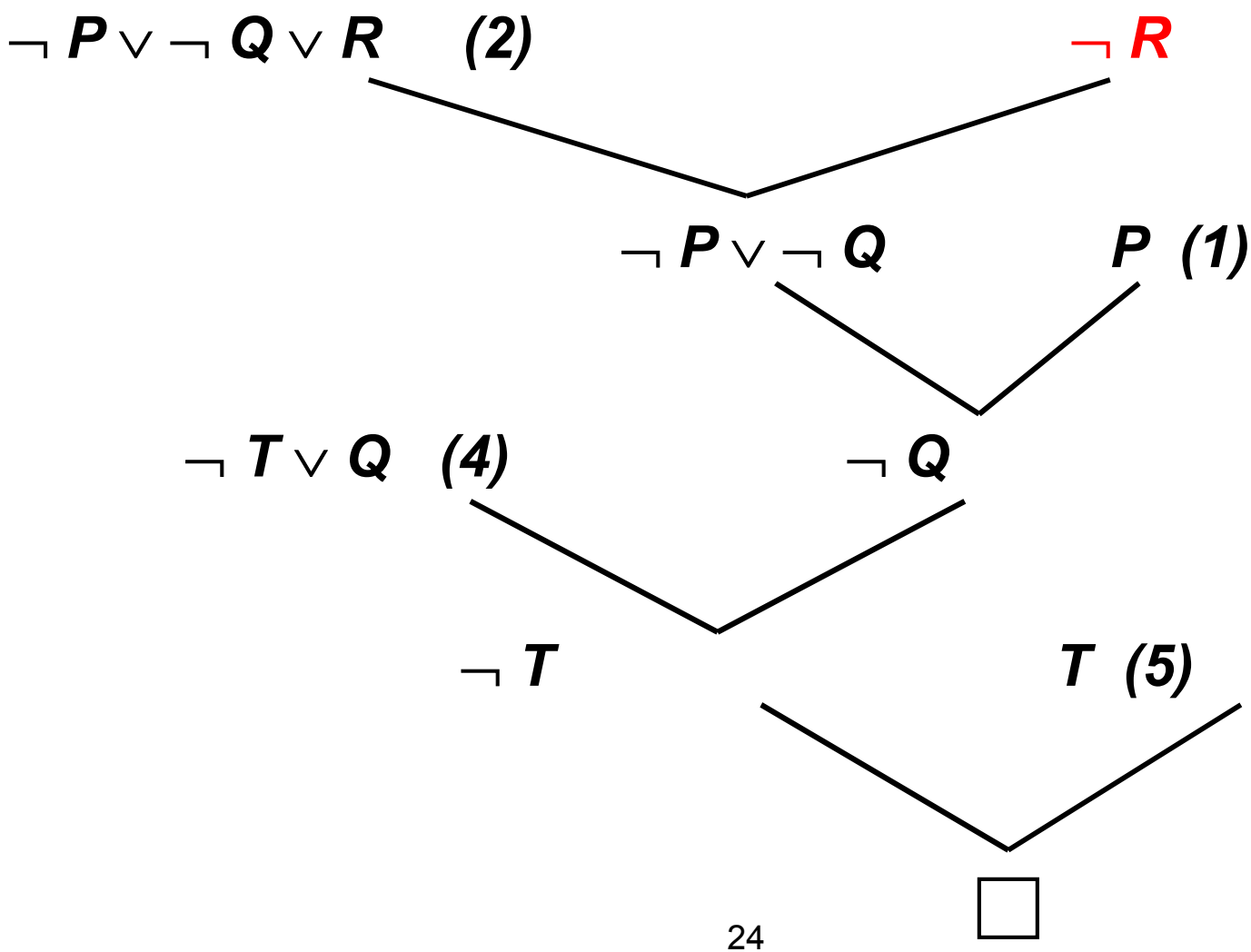
1. Put the premises or axioms into *clause form.*
2. Add the negation of what is to be proved, in clause form, to the set of axioms.
3. *Resolve* these clauses together, producing new clauses that logically follow from them.
4. Produce a contradiction by generating the empty clause.
5. The substitutions used to produce the empty clause are those under which the opposite of the negated goal is true.

A Facts in Propositional Logic

Given Axioms	Clause Form	
P	P	(1)
$(P \wedge Q) \rightarrow R$	$\neg P \vee \neg Q \vee R$	(2)
$(S \vee T) \rightarrow Q$	$\neg S \vee Q$	(3)
	$\neg T \vee Q$	(4)
T	T	(5)

Prove R!

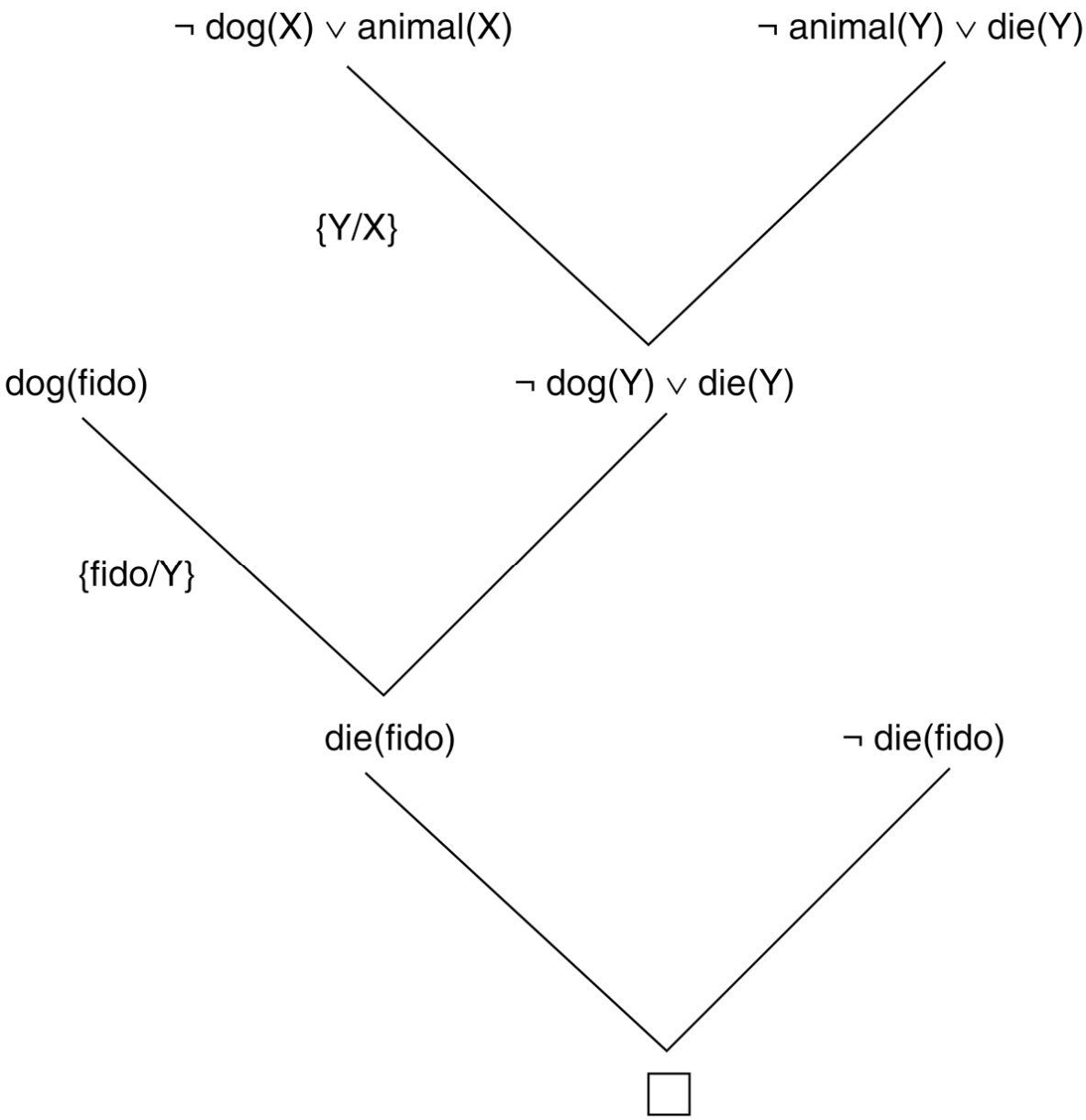
Resolution in Propositional Logic



Prove fido will die!

1. All dogs are animal.
2. Fido is a dog.
3. All animals will die.

Resolution proof for the “dead dog” problem.



Lucky Student

1. Anyone passing his history exams and winning the lottery is happy.
2. Anyone who studies or is lucky can pass all his exams.
3. John did not study but he is lucky.
4. Anyone who is lucky wins the lottery.

Prove that John is happy!

Anyone passing his history exams and winning the lottery is happy.

$\forall$ **X (pass (X,history)** $\wedge$ **win (X,lottery)** $\rightarrow$ **happy (X))**

Anyone who studies or is lucky can pass all his exams.

$\forall$ **X** $\forall$ **Y (study (X)** $\vee$ **lucky (X)** $\rightarrow$ **pass (X,Y))**

John did not study but he is lucky.

$\neg$ **study (john)** $\wedge$ **lucky (john)**

Anyone who is lucky wins the lottery.

$\forall$ **X (lucky (X)** $\rightarrow$ **win (X,lottery))**

1. $\neg$ **pass (X, history)** $\vee$ $\neg$ **win (X, lottery)** $\vee$ **happy (X)**
2. $\neg$ **study (Y)** $\vee$ **pass (Y, Z)**
3. $\neg$ **lucky (W)** $\vee$ **pass (W, V)**
4. $\neg$ **study (john)**
5. **lucky (john)**
6. $\neg$ **lucky (U)** $\vee$ **win (U, lottery)**

Into these clauses is entered, in clause form, the negation of the conclusion:

7. $\neg$ **happy (john)**

Anyone passing his history exams and winning the lottery is happy.

$\forall$ **X (pass (X,history) $\wedge$ win (X,lottery) $\rightarrow$ happy (X))**

Anyone who studies or is lucky can pass all his exams.

$\forall$ **X** $\forall$ **Y (study (X) $\vee$ lucky (X) $\rightarrow$ pass (X,Y))**

John did not study but he is lucky.

$\neg$ **study (john) $\wedge$ lucky (john)**

Anyone who is lucky wins the lottery.

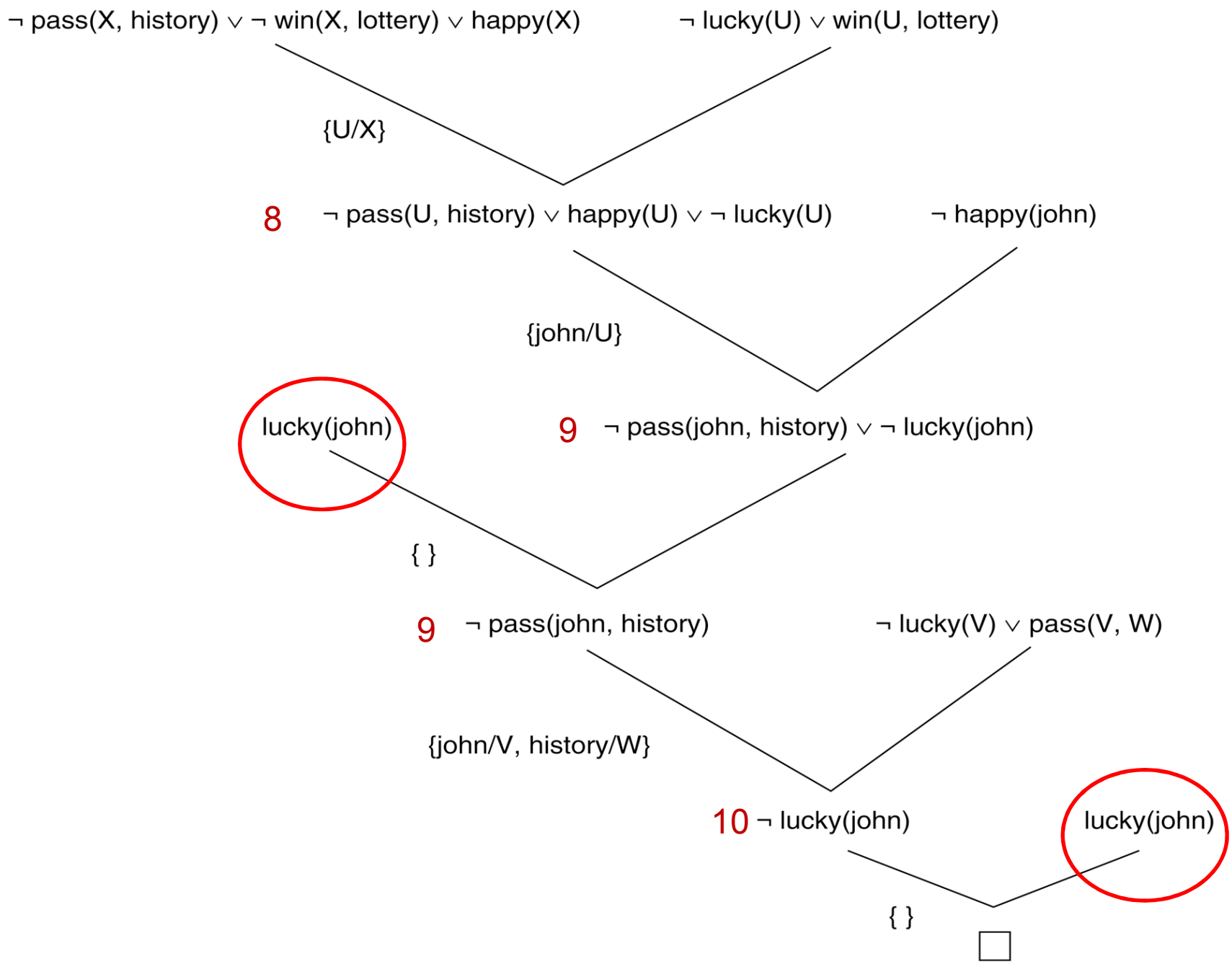
$\forall$ **X (lucky (X) $\rightarrow$ win (X,lottery))**

1. $\neg$ **pass (X, history) $\vee$ $\neg$ win (X, lottery) $\vee$ happy (X)**
2. $\neg$ **study (Y) $\vee$ pass (Y, Z)**
3. $\neg$ **lucky (W) $\vee$ pass (W, V)**
4. $\neg$ **study (john)**
5. **lucky (john)**
6. $\neg$ **lucky (U) $\vee$ win (U, lottery)**

Into these clauses is entered, in clause form, the negation of the conclusion:

7. $\neg$ **happy (john)**

One resolution refutation for the “happy student” problem.



Exciting life

1. All people that are not poor and are smart are happy.
2. Those people that read are not stupid.
3. John can read and is wealthy.
4. Happy people have exiting lives.

Can anyone be found with an exciting life?

All people who are not poor and are smart are happy. Those people who read are not stupid. John can read and is wealthy. Happy people have exciting lives. Can anyone be found with an exciting life?

We assume $\boxed{\forall \mathbf{X}\ (\mathbf{smart}\ (\mathbf{X}) \equiv \neg\ \mathbf{stupid}\ (\mathbf{X}))\ \mathbf{and}\ \forall \mathbf{Y}\ (\mathbf{wealthy}\ (\mathbf{Y}) \equiv \neg\ \mathbf{poor}\ (\mathbf{Y}))}$, **and get**:

$\forall \mathbf{X}\ (\neg\ \mathbf{poor}\ (\mathbf{X}) \wedge \mathbf{smart}\ (\mathbf{X}) \rightarrow \mathbf{happy}\ (\mathbf{X}))$

$\forall \mathbf{Y}\ (\mathbf{read}\ (\mathbf{Y}) \rightarrow \mathbf{smart}\ (\mathbf{Y}))$

$\mathbf{read}\ (\mathbf{john}) \wedge \neg\ \mathbf{poor}\ (\mathbf{john})$

$\forall \mathbf{Z}\ (\mathbf{happy}\ (\mathbf{Z}) \rightarrow \mathbf{exciting}\ (\mathbf{Z}))$

The negation of the conclusion is:

$\boxed{\neg\ \exists\ \mathbf{W}\ (\mathbf{exciting}\ (\mathbf{W}))}$

These predicate calculus expressions for the "exciting life" problem are transformed into the following clauses:

$\mathbf{poor}\ (\mathbf{X}) \vee \neg\ \mathbf{smart}\ (\mathbf{X}) \vee \mathbf{happy}\ (\mathbf{X})$ **1**

$\neg\ \mathbf{read}\ (\mathbf{Y}) \vee \mathbf{smart}\ (\mathbf{Y})$ **2**

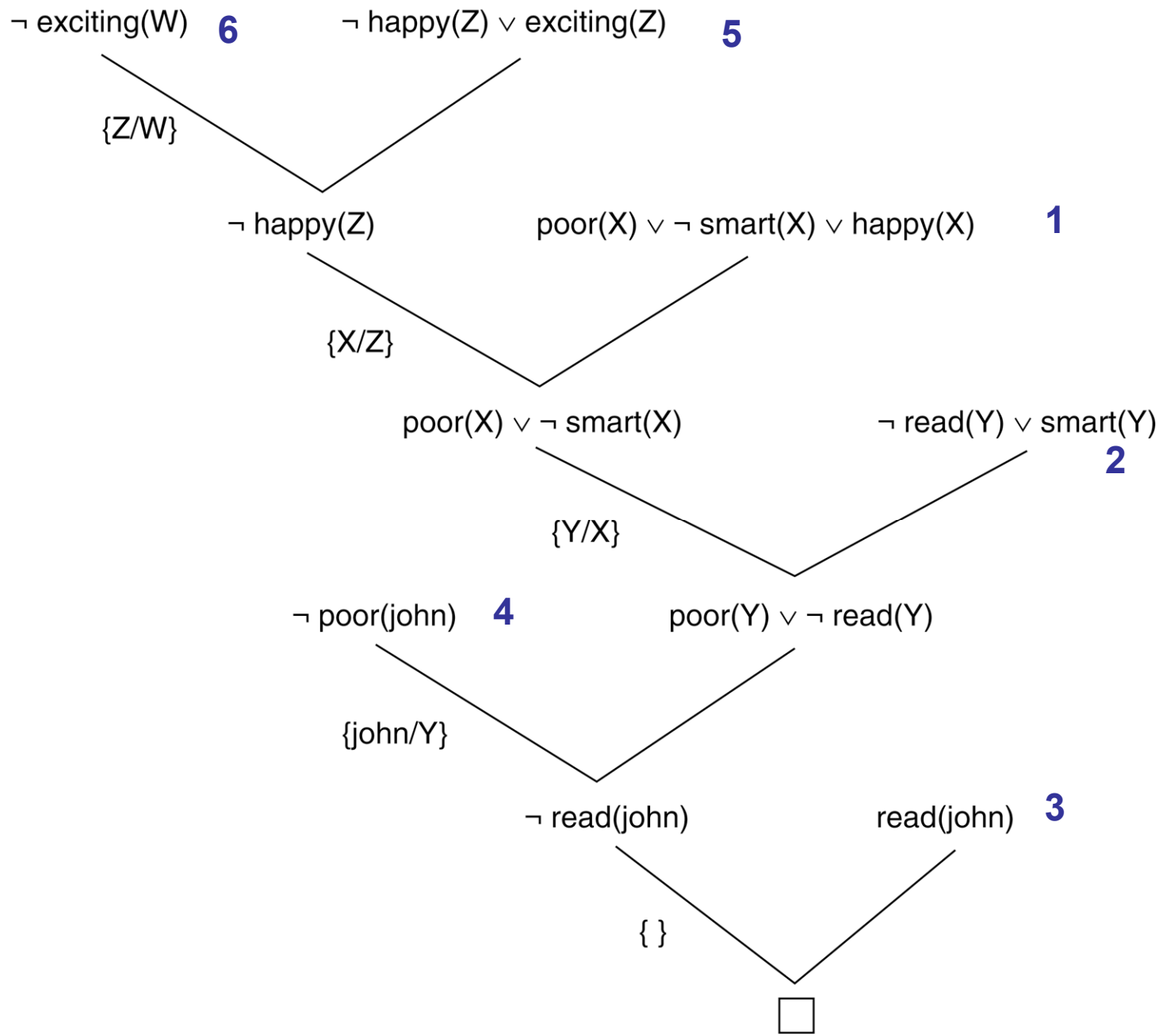
$\mathbf{read}\ (\mathbf{john})$ **3**

$\neg\ \mathbf{poor}\ (\mathbf{john})$ **4**

$\neg\ \mathbf{happy}\ (\mathbf{Z}) \vee \mathbf{exciting}\ (\mathbf{Z})$ **5**

$\neg\ \mathbf{exciting}\ (\mathbf{W})$ **6**

Resolution proof for the “exciting life” problem.



Resolution Example

1. **John likes all kinds of food.**

   $\forall x: food(x) \rightarrow likes(john, x)$

2. **Apples are food.**

   $food(apple)$

3. **Chicken is food.**

   $food(chicken)$

4. **Anything anyone eats and isn't killed by is food.**

   $\forall x: (\exists y: eats(y, x) \wedge \neg killedby(y, x)) \rightarrow food(x)$

5. **Bill eats peanuts and is still alive.**

   A. $eats(Bill, peanuts)$ B. $alive(Bill)$

6. **Sue eats everything Bill eats.**

   $\forall x: eats(Bill, x) \rightarrow eats(Sue, x)$

7. $\forall x: \forall y: alive(x) \rightarrow \neg killedby(x,y)$

Clause form:

1. $\neg$ *food(x1)* $\vee$ *likes(John, x1)*
2. *food(apples)*
3. *food(chicken)*
4. $\neg$ *eats(y4,x4)* $\vee$ *killedby(y4* , x4) $\vee$ food(x4)
5. *Eats (Bill, peanuts)*
6. *Alive (Bill)*
7. $\neg$ *eats(Bill,x7)* $\vee$ *eats(Sue,x7)*
8. $\neg$ *alive(x8)* $\vee$ $\neg$ *killedby(x8, y8)*

Resolution proof that John likes peanuts

¬ likes (John, peanuts) ¬ *food(x1)* ∨ *likes (John, x1)*

peanuts/x1

¬ *food(peanuts)* ¬ *eats(y4,x4)* ∨ *killedby(y4* . x4) ∨ food(x4)

peanuts/x4

¬ *eats(y4,peanuts)* ∨ *killedby(y4*. peanuts) *eats(Bill.peanuts)*

Bill/y4

killedby(Bill. peanuts) ¬ *alive(x8)* ∨ ¬ *killedby(x8 y8)*

Bill/x8, peanuts/y8

¬ *alive(Bill)* *alive(Bill)*

□